Tribhuvan University

Faculty of Humanities & Social Sciences

OFFICE OF THE DEAN

English I

BCA — Semester 1 — 2019 — Answer Sheet

Subject Code: CACS103

Group A

Attempt ALL the questions. [10 × 1 = 10]

1. i) Icon is defined as ……………
a) visual symbol used in a menu instead of natural language.
b) a device moved by hand to indicate position on the screen.
c) data programs
d) The set of software

ii) An ……. pen is one example of an input device.
a) electron b) electronic c) electronics d) electronically

iii) The word computational has ……………….
a) an adjective-forming suffix b) adverb-forming suffix
c) verb-forming suffix d) noun-forming suffix

iv) Newest is an example of ……………..
a) superlative adjective b) comparative adjective
c) absolute adjective d) absolute adverb

v) We were ………… to document our program.
a) instructor b) instructed c) instruct d) instruction

vi) The opposite meaning of 'preventing' is …………..
a) co-operating b) enabling c) reducing d) localizing

vii) Which of the following words has a destructive meaning?
a) cipher b) shield c) smart-card d) hacker

viii) The computer is ………… faster than the old one.
a) considered b) considerably c) considerable d) considering

ix) Our company is working on a new ………. of software products.
a) generation b) generative c) generated d) generate

x) The similar meaning (synonym) of the word inventive is ………………
a) Skilled b) creative c) awkward d) insufficient

[10 × 1 = 10]

Answer: Answers:
i) a) visual symbol used in a menu instead of natural language. — An icon is a small graphical representation of a program, file, or function in a GUI.

ii) b) electronic — An electronic pen (stylus) is an input device used with tablets and touchscreens.

iii) a) an adjective-forming suffix — The suffix "-al" forms adjectives from nouns (computation → computational).

iv) a) superlative adjective — "Newest" is the superlative form of "new," indicating the highest degree among three or more items.

v) b) instructed — "Instructed" is the past participle used in passive voice: "We were instructed..."

vi) b) enabling — "Preventing" means stopping something; its opposite is "enabling" (allowing or facilitating).

vii) d) hacker — A hacker is someone who breaks into computer systems, which is destructive. Cipher, shield, and smart-card are protective/neutral terms.

viii) b) considerably — "Considerably" is an adverb modifying the adjective "faster." Adverbs describe how, when, where, or to what extent.

ix) a) generation — "Generation" is a noun meaning a new version or stage of development.

x) b) creative — "Inventive" and "creative" are synonyms, both meaning able to produce new and original ideas.

Group B

Attempt any SIX questions. [6 × 5 = 30]

5. What are the major functions of the decision support system? Explain.

[5]

Answer: Decision Support System (DSS):
A Decision Support System is a computer-based information system that supports business and organizational decision-making activities. It analyzes large volumes of data and presents information in a way that helps managers make informed decisions.

Major Functions of DSS:

1. Data Management: DSS collects, stores, and organizes large amounts of data from internal and external sources. It provides tools for data retrieval, querying, and analysis.

2. Model Management: DSS uses mathematical and statistical models (optimization, simulation, forecasting) to analyze data and predict outcomes. Users can test different scenarios.

3. User Interface: Provides an interactive, user-friendly interface (dashboards, reports, visualizations) that allows decision-makers to interact with data and models easily.

4. Scenario Analysis: DSS enables "what-if" analysis, allowing users to test different assumptions and see potential outcomes before making decisions.

5. Reporting and Visualization: Generates reports, charts, graphs, and dashboards that present complex data in an understandable format for quick decision-making.

6. Knowledge Management: Some advanced DSS incorporate AI and expert systems to provide recommendations based on historical data and business rules.

6. How has the micro-chip technology changed the world forever? Illustrate.

[5]

Answer: Impact of Microchip Technology:

Microchip (integrated circuit) technology has revolutionized virtually every aspect of modern life:

1. Computing Power: Microchips enabled the transition from room-sized computers to smartphones that fit in pockets. A modern smartphone has more computing power than all of NASA's computers during the Apollo moon landing.

2. Communication: Mobile phones, internet routers, satellite systems, and fiber-optic networks all depend on microchips. Global instant communication became possible.

3. Healthcare: Microchips power MRI machines, CT scanners, pacemakers, insulin pumps, and wearable health monitors, saving millions of lives.

4. Transportation: Modern vehicles contain hundreds of microchips controlling engines, navigation, safety systems (ABS, airbags), and autonomous driving features.

5. Entertainment: Streaming services, gaming consoles, digital cameras, and smart TVs all rely on microchip technology for processing and display.

6. Economy: Microchips created entirely new industries (software development, e-commerce, app development) and transformed existing ones (banking, manufacturing, agriculture).

7. Space Exploration: Advanced microchips enable spacecraft navigation, rover control on Mars, and satellite imaging of Earth.

In essence, microchip technology is the foundation of the digital age, making information processing ubiquitous and affordable.

7. Give the meaning of "network of networks" and then explain the main use of ALGOL.

[5]

Answer: "Network of Networks" (Internet):
The term "network of networks" refers to the Internet — a global system of interconnected computer networks that use standardized communication protocols (TCP/IP) to link devices worldwide. It connects millions of private, public, academic, business, and government networks, allowing them to communicate and share resources seamlessly.

Key Characteristics:
• No single organization owns or controls it
• Uses packet switching to route data efficiently
• Supports multiple services: email, web browsing, file transfer, video streaming
• Scales from local networks (LAN) to global connectivity

ALGOL (ALGOrithmic Language):
ALGOL was one of the first high-level programming languages, developed in the late 1950s. It was designed specifically for scientific and numerical computation.

Main Uses of ALGOL:
1. Scientific Computing: ALGOL was extensively used for mathematical and engineering calculations, including matrix operations, differential equations, and simulations.

2. Algorithm Description: ALGOL's clear, structured syntax made it ideal for expressing algorithms in textbooks and research papers. Many computer science concepts were first described using ALGOL notation.

3. Language Design Influence: ALGOL influenced the design of many later languages including Pascal, C, Ada, and Java. Its block structure, recursion, and strong typing concepts became standard features.

4. Compiler Development: ALGOL was a testbed for compiler design research, leading to advances in parsing techniques and code optimization.

8. Give some specific names of clipboard computers now available in the market and then explain the two different jobs of 'infector' and 'detonator'.

[5]

Answer: Clipboard Computers (Tablets/2-in-1 Convertibles):
Popular models available in the market include:
• Apple iPad Pro (with Magic Keyboard)
• Microsoft Surface Pro
• Samsung Galaxy Tab S9 Ultra
• Lenovo Yoga Tab / ThinkPad X1 Fold
• Huawei MatePad Pro
• Amazon Fire HD tablets

Infector and Detonator (Computer Virus Components):

Many computer viruses consist of two main functional parts that work together:

1. Infector (Replication Component):
The infector is the part of the virus responsible for spreading and replicating itself. Its jobs include:
• **Self-replication:** Creates copies of the virus and attaches them to other executable files, documents, or boot sectors.
• **Propagation:** Spreads the virus to other computers through networks, email attachments, USB drives, or shared files.
• **Stealth:** Hides the virus from detection by disguising itself as legitimate files or modifying system tools.
• **Polymorphism:** Changes its code signature with each infection to evade antivirus signature detection.

2. Detonator (Payload Component):
The detonator (also called payload) is the part that executes the malicious action when triggered. Its jobs include:
• **Trigger activation:** Waits for specific conditions (date, time, number of infections, user action) before activating.
• **Damage execution:** Carries out the harmful action such as deleting files, corrupting data, stealing passwords, or displaying unwanted messages.
• **System disruption:** May crash systems, slow down performance, or create backdoors for hackers.
• **Information theft:** Some payloads steal sensitive data like banking credentials, personal information, or corporate secrets.

9. Why did the developers of the PAL system invent interlaced video? What are its advantages and disadvantages? Explain.

[5]

Answer: PAL (Phase Alternating Line) System:
PAL is an analog television color encoding system used in most of Europe, Asia, Africa, and Australia. It was developed in Germany in the 1960s.

Why Interlaced Video Was Invented:
Interlaced video was invented primarily to **reduce bandwidth requirements** while maintaining acceptable picture quality. In the early days of television, bandwidth was limited and expensive. By dividing each frame into two fields (odd and even lines), broadcasters could transmit video using half the bandwidth per field while maintaining the same frame rate, creating the illusion of smooth motion.

How Interlacing Works:
Each frame (complete image) is split into two fields:
• **Field 1:** All odd-numbered lines (1, 3, 5, ...)
• **Field 2:** All even-numbered lines (2, 4, 6, ...)
Fields are displayed alternately at twice the frame rate (e.g., 50 fields/sec for 25 frames/sec in PAL).

Advantages:
1. Bandwidth Efficiency: Requires approximately half the bandwidth of progressive scanning for the same perceived frame rate.
2. Reduced Flicker: Higher field rate (50Hz) reduces visible flicker compared to lower frame rates.
3. Smoother Motion: Better perception of motion due to higher temporal sampling rate.
4. Cost Effective: Allowed early TV systems to work with limited technology and transmission capacity.

Disadvantages:
1. Interlace Artifacts: Fast-moving objects show "combing" or "tearing" artifacts where edges from different fields don't align.
2. Reduced Vertical Resolution: Effective vertical resolution is lower than the stated number of lines.
3. Problems with Modern Displays: LCD and LED screens are inherently progressive, requiring de-interlacing which can cause quality loss.
4. Not Suitable for Computer Graphics: Text and sharp horizontal lines can flicker or shimmer in interlaced video.

10. "Computers are about to take people to places they have never been able to visit before." Explain the statement basing on the essay 'Fancy a fantasy Spacecraft?'.

[5]

Answer: Explanation of the Statement:

The statement from the essay "Fancy a Fantasy Spacecraft?" reflects the transformative power of computer technology and virtual reality in expanding human experiences beyond physical limitations.

1. Virtual Reality (VR) Exploration:
Computers enable immersive virtual reality experiences that transport users to places they cannot physically visit. VR headsets and simulations allow people to:
• Walk on the surface of Mars
• Dive into the depths of the ocean
• Explore ancient civilizations like Egypt or Rome
• Visit dangerous or inaccessible locations (volcanoes, radiation zones)

2. Space Exploration:
Computer simulations and robotics allow humans to explore space without leaving Earth. NASA uses computer-controlled rovers (Spirit, Opportunity, Curiosity, Perseverance) to explore Mars. Astronauts train extensively in computer simulations before actual missions.

3. Historical and Cultural Immersion:
Computer-generated reconstructions let people visit historical sites that no longer exist or have been damaged. Examples include virtual tours of the Titanic, ancient Pompeii, or the original Seven Wonders of the World.

4. Medical and Scientific Visualization:
Computers take scientists inside the human body (through 3D medical imaging), inside molecules (drug design), and inside weather systems (climate modeling) — places impossible to visit physically.

5. Gaming and Entertainment:
Video games create fantastical worlds (Middle-earth, Hogwarts, alien planets) that exist only in computer code but feel real to players, offering experiences beyond physical reality.

In essence, computers act as spacecraft of the mind, breaking the barriers of time, space, cost, and physical capability.

11. Discuss some of the tasks/jobs suited to robots only, and show the impact of robotics revolution felt in modern society.

[5]

Answer: Tasks Suited to Robots:

1. Repetitive Manufacturing: Assembly line work, welding, painting, and quality inspection in factories. Robots perform these tasks with consistent precision 24/7 without fatigue.

2. Hazardous Environments: Nuclear plant inspection, bomb disposal, deep-sea exploration, mining, and handling toxic chemicals. Robots protect human lives in dangerous conditions.

3. Precision Surgery: Robotic surgical systems (da Vinci) perform minimally invasive procedures with sub-millimeter accuracy, reducing patient trauma and recovery time.

4. Space Exploration: Robotic rovers, landers, and probes explore planets and moons where humans cannot yet travel. Examples: Mars rovers, Voyager probes.

5. Warehouse Automation: Autonomous robots in warehouses (Amazon Kiva) move inventory, sort packages, and optimize logistics with superhuman speed and accuracy.

Impact of Robotics Revolution:

1. Economic Transformation: Increased productivity and reduced labor costs, but also displacement of some manual jobs. New job categories in robot programming, maintenance, and AI development have emerged.

2. Healthcare Advancement: Surgical robots, rehabilitation exoskeletons, and elderly care robots improve patient outcomes and address healthcare worker shortages.

3. Quality of Life: Home robots (vacuum cleaners, lawn mowers) free people from mundane chores. Assistive robots help disabled and elderly people live independently.

4. Safety Improvement: Robots handle dangerous jobs, significantly reducing workplace injuries and fatalities in mining, construction, and manufacturing.

5. Ethical and Social Challenges: Concerns about job displacement, privacy (surveillance robots), autonomous weapons, and the need for new regulations and social safety nets.

Group C

Attempt any TWO questions. [2 × 10 = 20]

12. Perhaps you manage a computing specializing in multi-media hardware and software. Prepare a leaflet to inform companies of the potential benefits of using multi media.
OR
Write a letter of inquiry to a college or university requesting information about a degree program. Be specific in your request, and follow the criteria for writing letters of inquiry.

[10]

Answer: Option 1: Leaflet on Multimedia Benefits

LEAFLET: POWER YOUR BUSINESS WITH MULTIMEDIA SOLUTIONS

Why Multimedia?
In today's digital world, multimedia — the integration of text, graphics, audio, video, and animation — is essential for effective communication, marketing, and training.

Benefits for Your Company:

1. Enhanced Communication: Multimedia presentations convey complex ideas quickly and memorably. Studies show people remember 65% of visual content vs. 10% of text alone.

2. Increased Engagement: Interactive multimedia (videos, animations, infographics) captures audience attention and increases customer engagement on websites and social media.

3. Effective Training: Multimedia-based training modules improve employee learning outcomes by 60% compared to traditional methods. Employees can learn at their own pace with video tutorials and simulations.

4. Marketing Advantage: Video marketing generates 1200% more shares than text and images combined. Product demos, testimonials, and animated explainers boost conversion rates.

5. Cost Savings: Virtual meetings, webinars, and e-learning reduce travel and training costs while reaching global audiences instantly.

Our Services:
✓ Corporate video production
✓ Interactive e-learning modules
✓ 3D animation and product visualization
✓ Website multimedia integration
✓ Virtual reality experiences

Contact us today to transform your business communication!

Option 2: Letter of Inquiry

Your Name Your Address City, Postal Code Email: yourname@email.com Phone: 98XXXXXXXX Date: [Current Date] The Admissions Office Tribhuvan University Kirtipur, Kathmandu Subject: Inquiry about Bachelor in Computer Applications (BCA) Program Dear Sir/Madam, I am writing to request detailed information about the Bachelor in Computer Applications (BCA) program offered at your esteemed university. I have recently completed my +2 level in Science stream and am keenly interested in pursuing a career in information technology. I would appreciate it if you could provide the following information: 1. Admission Requirements: What are the academic qualifications, entrance exam details, and minimum GPA required for admission? 2. Course Duration and Structure: How many semesters does the program span? What is the credit system, and what are the core subjects in each semester? 3. Fees and Scholarships: What is the total tuition fee per semester? Are there any scholarships, grants, or financial aid options available for deserving students? 4. Facilities and Infrastructure: Does the university provide computer labs with modern equipment, high-speed internet, library access, and internship opportunities? 5. Career Prospects: What is the track record of campus placements? Do you have industry partnerships for practical training? 6. Application Deadline: What is the last date for submitting the application for the upcoming academic session? I would be grateful if you could send me a prospectus and application form at your earliest convenience. I am eager to join your program and build a strong foundation in computer applications. Thank you for your time and assistance. I look forward to your favorable response. Yours sincerely, [Your Signature] Your Name

13. You work in the purchasing department of an industry where a major project is being introduced at work. Employees working under your supervision need suggestion from you. Write a memo providing your response in regard to their individual responsibilities.
OR
Write a complete CV of your own to be sent with the covering letter while applying for a job.

[10]

Answer: Option 1: Memo

MEMORANDUM To: All Purchasing Department Staff From: [Your Name], Purchasing Manager Date: [Date] Subject: Individual Responsibilities for Upcoming Project Phoenix This memo outlines your specific responsibilities as we prepare for the launch of Project Phoenix, our major new manufacturing initiative effective next month. Ram Shrestha – Vendor Relations • Contact and evaluate at least three suppliers for raw materials • Negotiate payment terms and delivery schedules • Maintain vendor database and performance ratings Sita Gurung – Inventory Control • Conduct weekly stock audits of existing inventory • Coordinate with warehouse for storage allocation • Prepare requisition forms for project-specific materials Hari Prasad – Documentation and Compliance • Ensure all purchase orders are properly authorized • Maintain files for GST invoices and delivery challans • Track order status and report delays immediately General Instructions: • Daily stand-up meetings at 9:00 AM • Weekly progress report every Friday • All POs must be approved 48 hours before placement Your cooperation and diligence will ensure Project Phoenix launches on schedule and within budget. Please direct any questions to my office. [Your Name] Purchasing Manager

Option 2: Curriculum Vitae (CV)

CURRICULUM VITAE PERSONAL INFORMATION Name: Ram Bahadur Thapa Address: 123 Main Street, Kathmandu, Nepal Phone: +977-98XXXXXXXX Email: ram.thapa@email.com Date of Birth: January 15, 2000 Nationality: Nepali CAREER OBJECTIVE To secure a challenging position as a Software Developer where I can utilize my programming skills, problem-solving abilities, and passion for technology to contribute to organizational growth while advancing my professional career. EDUCATION 2021 – 2024 Bachelor in Computer Applications (BCA) Tribhuvan University, Kathmandu Percentage: 78% 2019 – 2021 +2 Science (Computer Science) St. Xavier's College, Kathmandu Percentage: 82% 2019 SLC (Secondary Education Examination) Kathmandu Model School GPA: 3.6 TECHNICAL SKILLS • Programming: Java, Python, C, JavaScript • Web Development: HTML, CSS, React, Node.js • Databases: MySQL, MongoDB • Tools: Git, VS Code, Eclipse, Figma • Operating Systems: Windows, Linux PROJECTS • E-Commerce Website: Built a full-stack online store using MERN stack • Student Management System: Desktop application in Java with MySQL • Portfolio Website: Responsive personal portfolio using React INTERNSHIPS Jun 2023 – Aug 2023 Software Development Intern TechNepal Pvt. Ltd., Kathmandu - Developed REST APIs for mobile application - Fixed 25+ bugs in production codebase CERTIFICATIONS • Oracle Certified Associate, Java SE 8 Programmer • Responsive Web Design – freeCodeCamp • Python for Data Science – Coursera EXTRA-CURRICULAR ACTIVITIES • Member, Computer Science Club (2022-2024) • Winner, Inter-College Coding Competition 2023 • Volunteer, Digital Literacy Camp for Rural Schools LANGUAGES • Nepali (Native) • English (Fluent) • Hindi (Conversational) REFERENCES Available upon request.

14. Write an essay about a problem that directly involves you. Choose a problem you see in your neighborhood, your college, or your job, and explain how it should be solved.

[10]

Answer: Essay: The Problem of Digital Illiteracy in Our College

In today's technology-driven world, digital literacy has become as fundamental as reading and writing. However, in my college, I have observed a significant gap in digital skills among students, particularly those from rural backgrounds who have limited prior exposure to computers. This problem affects their academic performance, confidence, and future employability.

The Problem:

Many students in my college struggle with basic computer operations. They face difficulties in typing assignments, creating presentations, using email, and navigating learning management systems. When professors assign online research or digital projects, these students fall behind. The digital divide creates an unequal learning environment where some students excel while others struggle with the medium rather than the content.

Consequences:

Academically, these students submit poorly formatted assignments, miss online deadlines, and cannot participate effectively in digital group discussions. Their grades suffer not because of poor understanding of subjects, but because of poor technical skills. Socially, they feel isolated and anxious when peers discuss technology comfortably. Professionally, they remain unprepared for modern workplaces where basic digital proficiency is assumed.

Proposed Solutions:

1. Mandatory Digital Literacy Course: The college should introduce a compulsory foundational course in the first semester covering MS Office, internet usage, email etiquette, and basic troubleshooting. This would level the playing field for all students.

2. Peer Mentoring Program: Tech-savvy students can volunteer as mentors, conducting weekly workshops and one-on-one sessions. Peer learning reduces intimidation and builds a supportive community.

3. Extended Computer Lab Hours: The college should keep computer labs open beyond class hours, including weekends, so students can practice at their own pace without time pressure.

4. Online Resource Portal: Create a college portal with video tutorials, FAQs, and practice exercises in both English and Nepali languages for self-paced learning.

5. Faculty Training: Ensure that faculty members are patient and equipped to assist students with varying digital skill levels, rather than assuming universal proficiency.

Conclusion:

Digital illiteracy is not an individual failure but a systemic gap that colleges must address proactively. By implementing these solutions, my college can ensure that every student, regardless of background, has the digital skills necessary to succeed academically and professionally in the 21st century.

Note: These answers are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.